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CENTRAL SEROTONIN FUNCTIONS AND AVAILABILITY OF TRYPTOPHAN TO THE BRAIN

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ABSTRACT

Tryptophan (TRP) was the first amino acid to be recognized initially as being essential for normal growth of young animals and later as an essential amino acid for human nutrition. It plays an important role in maintaining a large number of normal physiological functions. It is the precursor of several neuroactive compounds including serotonin (5-hydroxytryptamine 5-HT) which functions as a neurotransmitter in the brain and is involved in the regulation of a variety of physiological and behavioral functions. It is well established that TRP changes clearly influence brain 5-HT synthesis but how much this might affect the availability of serotonin to its receptors in order to produce a function is what has been analysed in the present article.

INTRODUCTION

Serotonin (5-hydroxytryptamine, 5-HT) first isolated from blood serum, was found to have constrictive effect on blood vessels. It was identified as a neurotransmitter in the mammalian brain in 1950's and its anatomy was elucidated (Dahlstrom and Fuxe, 1965). In the brain, 5-HT containing neurons whose cell bodies are confined to the raphe nuclei in the mid and hindbrain send projections rostrally to many areas of the brain and caudally these neurons extend to the medulla and spinal cord. It is because of this wide projection that 5-HT neurons have the potential to influence large areas of the brain and spinal cord functions. Serotonin in the brain is synthesized from essential amino acid L-tryptophan (TRP). The principle metabolite of 5-HT degradation is 5-hydroxyindole acetic acid (5-HIAA) (Fig. 1).

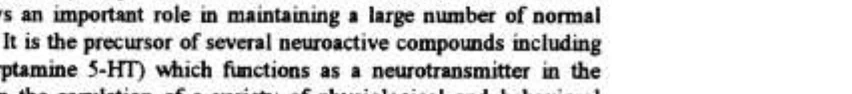

Fig.1. Synthesis and degradation of serotonin.
TH: Tryptophan hydroxylase, AAAD: Amino acid decarboxylase, MAO: Monoamine oxidase.

Interactions between serotonergic subsystems and between 5-HT and other neurotransmitters make it possible for 5-HT to contribute to the regulation of many important psychobiological and physiological functions (Fig. 2). The many functions in which 5-HT participates are mediated by a variety of receptors. These 5-HT receptors differ with respect to distribution in the brain, structure, location on the (cell body or dendritic) or postsynaptic neuron, pharmacology and second-messenger system. According to the latest classification, atleast 7 distinct classes of 5-HT receptors have been identified in the central nervous system (CNS) of mammals (Peroutka, 1995).

Factors affecting the availability of tryptophan to the brain and 5-HT synthesis.

The role of TRP in controlling brain 5-HT synthesis is well established (Ashcroft, 1965). Under physiological conditions, the rate of 5-HT synthesis is mainly dependent on the concentration of its precursor TRP in the brain. The brain tryptophan concentration is in turn determined by certain factors like, plasma free and total (Tagliamonte et al, 1973) TRP concentration; plasma concentration of other large neutral amino acids (LNAAs) which compete with TRP for uptake into the brain (Fernstrom, 1983). Physiological factors like food ingestion, starvation, stress, exercise etc. alter the availability of tryptophan to the brain by changing above parameters and hence 5-HT synthesis, metabolism and preseumably transmission. Rate of serotonin synthesis in the brain is sensitive to changes in the concentration of its precursor TRP because the enzyme TRP hydroxylase, which is confined to the serotonergic neurons (Knowles et al, 1984) and catalyzes the rate limiting step of 5-HT synthesis, is not saturated with its substrate at normal brain TRP concentration (Carlsson and Lindqvist, 1978) and hence lowering or increasing the availability of TRP to the brain decreases or increases serotonin synthesis.

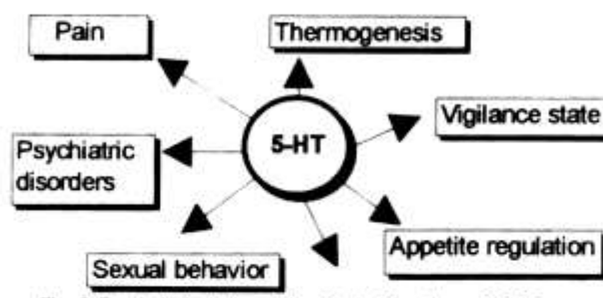

Fig. 2. Psychobiological and physiological functions of 5-HT

Availability of tryptophan: 5-HT synthesis and release.

Amino acid L-TRP is the precursor of neurotransmitter serotonin, which is involved in controlling various vital body functions. It is now well established that brain serotonin synthesis is influenced by the availability of its precursor, which cannot be synthesized within the body and is obtained exclusively through the diet. Procedures that increase or decrease the concerntration of TRP within the brain produce corresponding changes in 5-HT concentration in whole brain. Regardless of the mechanism by which TRP increases the levels of 5-HT in the brain, injection of the amino acid almost always increases 5-HT synthesis. The question arises as to what is the fate of this newly synthesized 5-HT and whether or not this 5-HT produced by TRP has a functional consequence? In order to fulfill its role as a neurotransmitter the 5-HT must be released from the presynaptic neuron into the synapse and interact with one of the many 5-HT receptors . Although increased availability of TRP to the brain increases brain 5-HT synthesis, the extent to which TRP induced increases of 5-HT actually cause enhanced 5-HT neurotransmission is still a matter of debate. A number of studies have been performed to measure the TRP induced synthesis and release of 5-HT. Studies have shown that various drugs (Bel and Artigas, 1996) and other physiological conditions (Leibowitz, 1988) alter the availability of TRP to the brain resulting in either a decrease or an increase in 5-HT metabolism. Brain tryptophan concentration shows typical diurnal variations (Delgado et al, 1989). Parallel changes occur in 5-HT metabolism.

Westerink and Vries, (1991) demonstrated that 100 mg/kg systemic dose of L-TRP stimulated the formation of 5-HTP in the striatum about 3 fold. This stimulation led to an increase in the release of 5-HT that was 150% of the control values as measured by extracellular level of 5-HT. Ih a similar study by Carboni et al, (1989), a 2 fold increase in the 5HT content of cortical dialysate after a similar dose of TRP was observed. Other workers however have been unable to observe significant increase in the dialysate levels of 5-HT after TRP treatment (Sleight et al, 1988). Various findings suggest that release of 5-HT depends on the neuronal firing rate and 5-HT synthesis is particularly sensitive to TRP supply in conditions when neurons are active. Peripheral TRP loading in food deprived rats increased 5-HT release in the lateral hypothalamic region which is known to be involved in food intake (Schwartz et al, 1990). Electrical stimulation of different raphe nuclei has also shown to stimulate 5-HT release. This was demonstrated in a study where the electrically evoked release of 5-HT in the hippocampus of rats was markedly enhanced by pre-treament with L-TRP (Sharp et al, 1992). In another similar study pre-treament with 8-OH DPAT attenuated 5-HT synthesis and release in the hippocampus of rats (Fernstrom et al, 1990). 8-OH DPAT acts by stimulating 5-HT1A receptors which turn off the neuron thus decreasing the synthesis and release of neurotransmitter (Fernstrom et al, 1990).

TRP availability and feeding behavior.

Brain 5-HT is known to be involved in the control of eating behaviour (Leibowitz, 1990) and disturbances in serotonergic function lead to eating disorders. The hypophagic function of 5-HT is mediated through 5-HT2C receptors. In a recent study hyperphagia was observed in mutant mice lacking 5-HT2C receptors (Laurence et al, 1995) suggesting the hypophagic role of these receptors. A number of studies have examined the effect of the availability of TRP (precursor of 5-HT) on brain 5-HT synthesis and release. TRP is an essential amino acid. Ingestion of food is one of the most potent physiological process that alters the blood concentration of TRP and its LNAA competitors. By this mechanism, food ingestion influences serotonin synthesis in brain. Rat brain TRP and 5-HT synthesis has long been known to increase after large carbohydrate-low protein meal due to fall in plasma concentration of amino acids which compete with the transport of TRP to the brain (Crandall et al, 1980). This was also confirmed by Latham and Blundell (Latham and Blundell, 1979) who observed a reduction in food intake of freely feeding rats after TRP administration as increased 5-HT function is known to reduce food intake. Weinberger et al, (1978), on the other hand found no acute effect of TRP administration on food intake in fasted rats. In a recent study, a lack of effect of TRP on carbohydrate intake was also observed in humans (Brewerton et al, 1994). Evidence also suggests that TRP influences food choice (Latham and Blundell, 1979). Administration of TRP within meals decreased carbohydrate and increased protein intake. Conversly, if plasma TRP was decreased by giving a TRP deficient amino acid mixture the protein intake was decreased (Fernstrom, 1990). In some hypothesis of macronutrient appetite regulation, brain TRP concentration and serotonin synthesis are viewed as fluctuating from meal to meal as a function of the protein and carbohydrate contents of the meal. Teff et al, (1989) conclude from their studies that protein and carbohydrate meals fail to alter human brain TRP concentration sufficiently enough to influence serotonin synthesis. Fernstrom and Fernstrom (Fernstrom and Fernstrom, 1995) on the other hand demonstrated that rat brain TRP concentration and serotonin synthesis are responsive to sequential ingestion of protein and carbohydrate meals if there is a sufficient interval between meals.

Tryptophan and eating disorders.

A number of behavioural, pharmacological and neurochemical studies in animals suggest that altered function of serotonin may play a role in the development of two well known eating disorders namely anorexia nervosa and bulimia nervosa (Anderson et al, 1990). Bulimia nervosa is characterised by episodes of binge eating followed by purging and vomitting while anorexia nervosa is a condition of self inflicted starvation or dieting. Various studies in patients with eating disorders support the fact that patients with bulimia nervosa and anorexia nervosa have signs of altered serotonergic activity (Goodwin et al, 1989). It is suggested that food restriction induced changes of brain 5-HT could provoke compensatory changes at postsynaptic hypophagic 5-HT receptors (Weltzin et al, 1994). Recently work done in our laboratory also demonstrated decreased 5-HT concentration and synthesis rate compared to freely feeding controls in the hypothalamus of rats fed on restricted feeding schedule of 4h/day for 5 days (Haleem and Haider,1996). Anorexia due to various diseases such as cancer, liver cirrhosis and uremia is recognized by an increase in the availability of TRP to the brain (Fanelli and Cangiano, 1991). This in turn leads to enhanced brain 5-HT synthesis resulting in reduced food intake. In a recent study increased plasma TRP and brain 5-HT which led to anorexia, was observed in tumor bearing rats. (Muscaritoli et al, 1996).

TRP and other Psychiatric disorders .

Depression/Aggression:

Considerable evidence has been accumulated to support the hypothesis that alteration in the serotonergic neuronal function in the CNS occur in patients with major depression (Meltzen , 1989). It is suggested that depression is due to a deficiency of available serotonin or sub-responsivity of serotonin receptors in relavent brain areas (Meltzen, 1989). Synthesis of serotonin is dependent on dietary intake of its precursor TRP (Ashcroft et al, 1965). Low plasma free and total TRP in depression has been reported by various scientist (Moller et al, 1990, Delgado et al, 1990), which normalized on recovery. Lower TRP/LNAA values which improved on treatment with 5-HT reuptake inhibitiors was observed by Moller et al, (1990). Recent studies suggest that a short term reduction in TRP intake causes a rapid lowering of mood in healthy humans and a rapid return of depressive symptoms in patients whose depression had remitted (Delgado et al, 1990). Cleare et al, (1994) demonstrated an enhancement of aggression on TRP depletion in subjects who had preexisting aggressive traits. These results strongly suggest that low availabilty of plasma TRP is involved in depression/aggression.

TRP and treatment of Psychiatric disorders.

Both increases and decreases in dietary TRP intake lead to corresponding changes in brain TRP and 5-HT levels in laboratory animals. TRP depletion has been shown to alter various behavioral functions which are reversed by TRP repletion probably through alteration in cerebral 5-HT function. Increasing the availability of TRP to the brain by TRP infusions may be useful in the treatment of psychiatric disorders in which reduced serotonergic activity is implicated. TRP is used as an antidepressant compound. In combination with other antidepressants it increases their treatment efficacy. Hopefully its use could be extended as an adjuvent in drug therapy which lead to depression. Bulmia nervosa is a disorder characterized by abnormal ingestion of large amounts of food, whereas anorexia nervosa is condition of self inflicted starvation. Neurochemical studies suggest altered serotonergic activity in bulimia nervosa and anorexia nervosa. A role of TRP availability in the precipitation of these eating disorders is less clear. Future studies may possibily help to develop tryptophan containing medicines as treatment for these psychological eating disorder

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